V4.2 Toy Model 3A
Origin of Complex Amplitudes and Hilbert-Space Structure

Rev A - Exploratory Mathematical Compatibility Report

Date: 2026.05.25 | *Continuation after Toy Model 2G*

Status: Toy Model 3A tests whether complex amplitudes can emerge from real hidden geometry after a 3T -> 1T projection. It does not prove that complex numbers are unavoidable in nature and does not derive the Born rule.

1. Purpose

Toy Model 2 closed the quantum-side compatibility track using standard complex amplitudes. Toy Model 3A begins the next layer: asking whether the complex amplitude structure itself can be represented as the shadow of real hidden temporal geometry.

The narrow target is not to derive all of Hilbert space from first principles. The target is to show that once a 3T hidden orientation is restricted to a 1T observable axis, the two transverse hidden temporal degrees of freedom naturally form a real phase plane. A real phase plane with a 90-degree rotation operator is mathematically equivalent to complex numbers.

2. Starting from real geometry only

Begin with a real three-dimensional hidden temporal space:

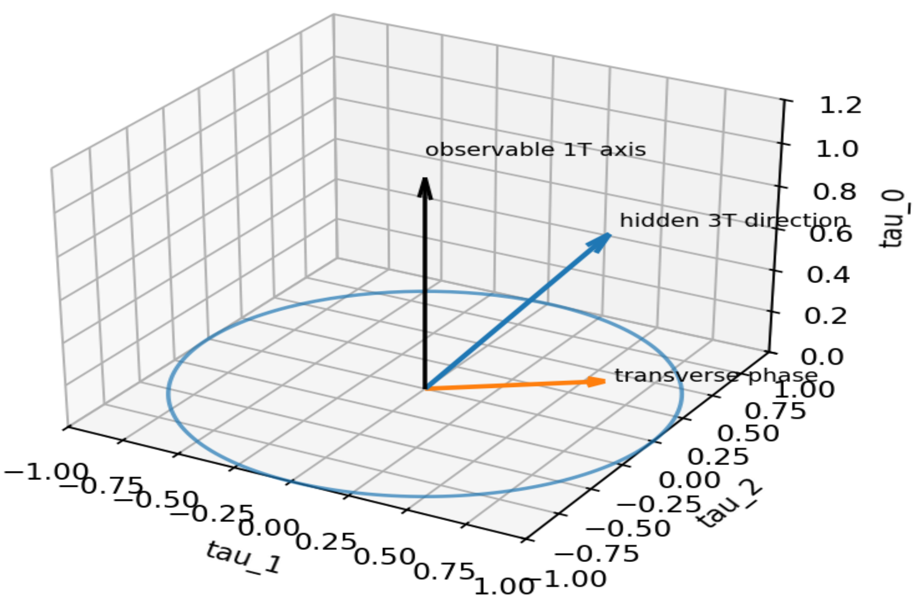
```
T_hidden = span(tau_0, tau_1, tau_2)
```

The observable 1T projection selects one axis, tau_0. The two remaining hidden temporal directions form a transverse plane:

```
P_transverse = span(tau_1, tau_2)
```

The key point: no complex number is inserted at this stage. We only have a real two-dimensional plane.

3T -> 1T projection leaves a transverse 2D phase plane



3. The real rotation operator J

On the real transverse plane, define J as a 90-degree rotation:

```
J = [[0, -1],
     [1,  0]]
```

Direct calculation gives:

```
J^2 = -I
```

This is the first structural bridge to the imaginary unit. In ordinary complex arithmetic, multiplying by i also means a 90-degree rotation and satisfies i^2 = -1.

Important guardrail: this does not prove that nature must use complex numbers. It shows that a real hidden phase plane with orientation-preserving rotations has the same algebraic structure as complex numbers.

4. Phase rotation without starting with i

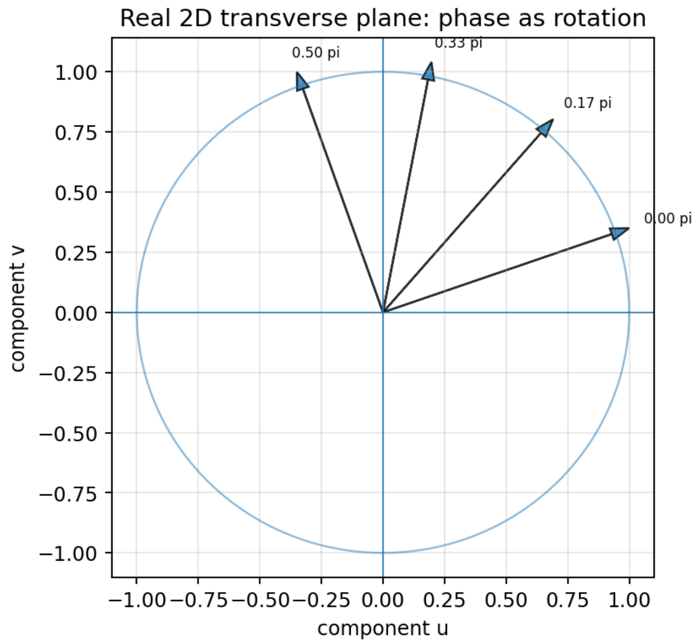
A rotation in the transverse real plane is:

```
R(phi) = cos(phi) I + sin(phi) J
```

This is a completely real 2x2 matrix. It preserves the squared length of vectors in the phase plane. Only after this real structure is defined can we use the shorthand:

```
(u, v) <-> z = u + i v
R(phi)(u, v) <-> exp(i phi) z
```

Plain meaning: the complex phase exp(i phi) is a compact notation for a real rotation in the two hidden transverse temporal degrees of freedom.



5. Squared length and probability weight

For a transverse vector a = (u, v), the invariant squared length is:

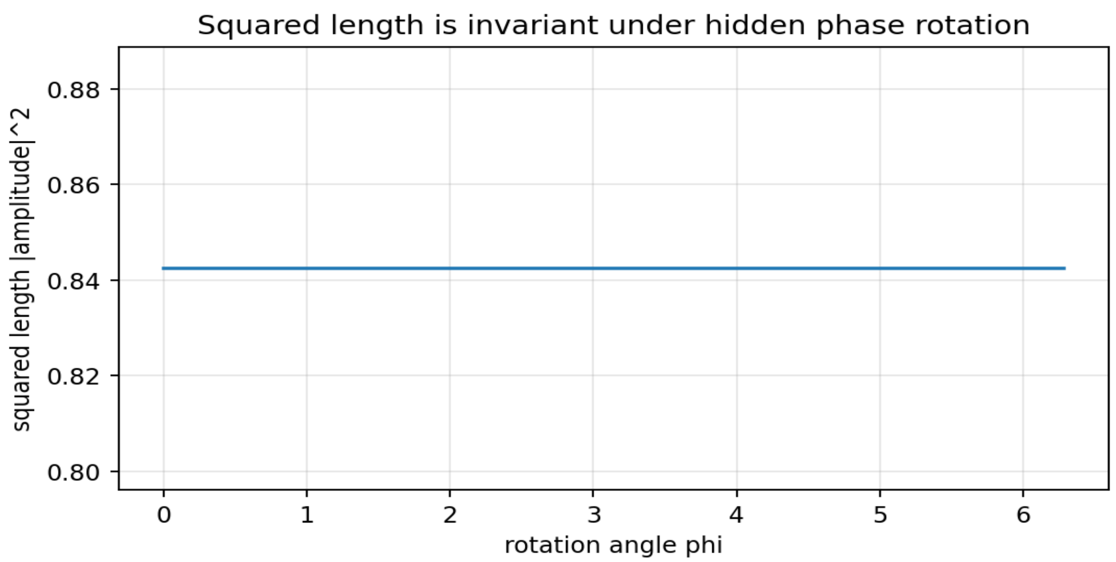
```
|a|^2 = u^2 + v^2
```

Under phase rotation, the squared length is unchanged:

```
|R(phi) a|^2 = |a|^2
```

This is the geometric origin of the squared-modulus structure used by complex amplitudes. If probability weight is required to be invariant under hidden phase rotation, then the natural quadratic candidate is u^2 + v^2.

Guardrail: this gives a geometric route to Born-rule form, but it is not a complete derivation of the Born rule. It assumes that probability weight is read from the invariant quadratic norm.



6. Finite-dimensional Hilbert structure

The transverse plane has two structures: a real Euclidean inner product g and a 90-degree rotation J. Together they define a complex-compatible inner product:

```
h(u, v) = g(u, v) + i g(Ju, v)
```

Term reminders:

• g(u,v) is the ordinary real dot product.

• J is the 90-degree rotation operator on the transverse plane.

• h(u,v) is the Hermitian inner product used in complex quantum mechanics.

• Hermitian means the inner product has the conjugation symmetry needed for complex amplitude spaces.

This shows how a real phase plane plus J can be repackaged as a complex one-dimensional Hilbert space. Multiple outcomes give a direct sum of such phase planes, which is equivalent to a finite-dimensional complex Hilbert space.

7. L2 Hilbert-space extension

For a continuous outcome variable x, assign a transverse phase-plane vector at each x:

```
psi(x) = (u(x), v(x))
```

The square-integrability condition is:

```
Integral [ u(x)^2 + v(x)^2 ] dx < infinity
```

Using the complex shorthand psi(x) = u(x) + i v(x), this becomes the usual complex L2 condition:

```
Integral |psi(x)|^2 dx < infinity
```

Therefore, the L2 Hilbert-space form can be recovered as a representation of real square-integrable fields valued in the hidden transverse phase plane.

8. Numerical checks

Check	Result
Random samples	10000
Max J^2 + I error	0.000e+00
Max rotation orthogonality error	2.220e-16
Max norm preservation error	3.553e-15
Max rotation composition error	3.671e-15
Max real-rotation / complex-phase equivalence error	4.441e-16
Max Hermitian inner-product invariance error	1.986e-15
Max qubit normalization error	4.441e-16
Max qubit phase-plane Born-form error	3.331e-16

9. Interpretation inside the 6D Projection Framework

Toy Model 3A supports the following limited interpretation:

```
3T hidden temporal geometry
      -> 1T observable projection axis
      -> two transverse hidden temporal degrees remain inaccessible
      -> transverse real phase plane carries rotations
      -> J^2 = -I
      -> complex phase notation becomes natural
      -> Hilbert-space amplitude structure can be represented
```

This is stronger than simply assuming complex amplitudes, because it identifies a real geometric source for the imaginary direction: the two hidden temporal degrees that remain after the 1T projection.

10. What Toy Model 3A achieves

• It starts from real 3D hidden temporal geometry, not from complex numbers.

• It shows that a 3T -> 1T projection leaves a real two-dimensional transverse phase plane.

• It defines a real 90-degree rotation operator J satisfying J^2 = -I.

• It shows that complex phase exp(i phi) is equivalent to real rotation exp(phi J).

• It recovers squared-modulus structure as invariant squared length in the phase plane.

• It sketches how finite and L2 Hilbert structures can be represented from this real phase-plane geometry.

11. What Toy Model 3A does not achieve

• It does not prove that complex numbers are unavoidable in nature.

• It does not derive the Born rule from first principles.

• It does not derive the Schrodinger equation.

• It does not derive quantum field theory or the Standard Model.

• It does not prove the physical existence of 3T.

• It does not yet produce new experimental predictions.

12. Close-out statement for 3A

Toy Model 3A successfully constructs a real-geometric origin candidate for complex amplitude structure. A 3T hidden temporal space restricted to a 1T observable axis leaves a two-dimensional transverse phase plane. A real 90-degree rotation operator J on this plane satisfies J^2 = -I, making the plane algebraically equivalent to the complex numbers. Phase factors exp(i phi) can therefore be interpreted as real rotations in hidden transverse temporal geometry. The Hilbert-space norm and squared-modulus form arise as the invariant squared length of this phase-plane representation. This is a compatibility construction and geometric explanation candidate, not a full derivation of quantum mechanics or the Born rule.

13. Recommended next chapter

The next logical chapter is Toy Model 3B: Dynamics from Phase Rotation - Toward Schrodinger-Type Evolution.

1. Define phase evolution as uniform rotation in the hidden transverse phase plane.

2. Test whether norm-preserving first-order time evolution requires an anti-symmetric generator.

3. Translate the real generator J into the complex form involving i.

4. Check whether a Schrodinger-type equation emerges as the compact complex notation for real rotation.

5. Keep the guardrail: derive only structural compatibility, not the full physical Schrodinger equation for all systems.

14. Rebind prompt for continuation

```
We are resuming after Toy Model 3A.

3A tested the origin of complex amplitudes from reciprocal projection.

Starting point:
T_hidden = span(tau_0, tau_1, tau_2)

1T observable projection selects tau_0.
The transverse hidden phase plane is:
P_transverse = span(tau_1, tau_2)

Define real 90-degree rotation:
J = [[0, -1],
     [1,  0]]

Then:
J^2 = -I

Real phase rotation:
R(phi) = cos(phi) I + sin(phi) J

Complex shorthand:
(u, v) <-> z = u + i v
R(phi)(u,v) <-> exp(i phi) z

Result:
Complex amplitudes can be represented as real hidden transverse phase-plane vectors.
The squared modulus |z|^2 is the invariant squared length u^2 + v^2.

Guardrail:
3A does not prove complex numbers are unavoidable, does not derive Born rule,
and does not prove physical 3T existence.

Next recommended chapter:
Toy Model 3B - Dynamics from Phase Rotation: toward Schrodinger-type evolution.
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